

Topics:

➤ Reviews and Configuration

1. Walkthroughs and Inspections
2. Software Configuration Management
3. Quality System Documentation

❖ Software Quality Assurance (SQA)

Software Quality Assurance is a systematic process to ensure software meets quality standards, is reliable, and works as expected.

- **Reviews**

Reviews involve evaluating software, documents, or processes to check for errors or defects. It ensure compliance with standards or regulations. It also improve quality and reliability.

- **Types of reviews:**

1. **Code Review:** Evaluating software code to ensure it's correct, efficient, and maintainable.
2. **Design Review:** Evaluating software design to ensure it meets requirements and is feasible.

- **Configuration**

Configuration involves setting up and arranging software, hardware, or documentation to work properly.

1. Walkthroughs and Inspections

What is a Walkthrough?

A **Walkthrough** is an informal meeting. It is a step-by-step guide or tour to help understand a process or system. The software developer explains the code, document, or design to a small group of people. In walkthrough, no formal process or report is required. Team members can ask questions and suggest improvements.

Purpose: To get feedback, find mistakes, and improve the work.

Example:

A developer shows their code to team members to check logic, style, and find any errors.

What is an Inspection?

An **Inspection** is a formal review process. It follows strict steps like planning, preparation, meeting, and reporting. A trained moderator leads the inspection. The focus is on finding defects early (like mistakes in design, code, or documents). Inspection generates written reports. **(OR)**

A formal evaluation of software, documents, or processes to identify errors or defects.

Example:

A software company checks a design document thoroughly before starting coding.

✓ Difference between Walkthrough and Inspection:

Feature	Walkthrough	Inspection
Formality	Informal	Formal
Leader	Developer	Trained Moderator
Report	Not required	Required
Focus	Understanding and improvement	Defect detection
Process	Flexible	Strict and structured

2. Software Configuration Management (SCM)

What is SCM?

Software Configuration Management (SCM) is a process to manage changes in the software to ensure consistency, quality and reliability. It helps to track, control, and record changes in code, documents, and other software items.

Main Goals of SCM:

- ✓ Keep software versions safe.
- ✓ Avoid mistakes due to untracked changes.
- ✓ Maintain history of all changes.
- ✓ Support team collaboration.

Key Activities in SCM:

1. Version Control:

Tracks different versions of software files (e.g., using Git).

2. Change Control:

Defines how a change will be proposed, approved, and implemented.

3. Configuration Identification:

Labels important items like documents, code files, and design files.

4. Configuration Status Accounting:

Keeps records of all configuration items and their changes.

5. Configuration Auditing:

Checks that all changes were done properly and nothing is missing.

Example Tools for SCM:

Git, GitHub, Bitbucket, SVN.

3. Quality System Documentation

What is Quality System Documentation?

It is the written record of how software quality will be planned, managed, and checked. It defines quality standards and procedures to be followed by the project team.

Main Components of Quality System Documentation:

1. **Quality Manual:**

High-level document that describes the overall quality policy of the organization.

2. **Quality Plan:**

Explains how quality will be maintained for a particular software project.

3. **Process Descriptions:**

Step-by-step details of each process like testing, reviews, coding standards.

4. **Work Instructions:**

Specific instructions for doing a task (like how to write unit test cases).

5. **Checklists and Templates:**

Predefined formats for consistency (like test case template, code review checklist).

6. **Records and Reports:**

Stores data like test results, review reports, and audit findings.

✓ Importance of Quality Documentation:

- ✓ Helps team members follow the same quality standards.
- ✓ Provides a reference for audits and reviews.
- ✓ Improves communication among teams.
- ✓ Supports continuous process improvement.

One-liner for Each

Topic	Short Summary
Walkthrough	Informal review for understanding and suggestions.
Inspection	Formal review to detect defects with reports.
SCM	Process to manage and control software changes.
Quality Documentation	Written guidelines and records for quality management.